

Home Bias and Risk Sharing

Sørensen et al. 2007, Empirical Replication and Extension
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Introduction

Panel-data Regressions

Introduction

Motivation Home Bias and Risk Sharing

- Macroeconomic Risk Sharing
 - international diversification and smoothing of income streams
 - hedging against idiosyncratic income shocks
- Home Bias
 - tendency of investors to hold a disproportionately large share of domestic assets
 - less internationally diversified portfolios than financial theory would predict

They do not need to go hand in hand...

Agents could also smooth consumption through borrowing and lending.

Panel-data Regressions

Panel-data Regressions

Income Risk Sharing Panel Regression

$$\Delta \log GNI_{it} - \Delta \log GNI_t = \text{constant} + \kappa (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it} \quad (1)$$

with

$$\kappa = \kappa_0 + \kappa_1(t - \bar{t}) + \kappa_2(EHB_{it} - \overline{EHB_t}) \quad (2)$$

Consumption Risk Sharing Panel Regression

$$\Delta \log C_{it} - \Delta \log C_t = \text{constant} + \eta (\Delta \log GDP_{it} - \Delta \log GDP_t) + \epsilon_{it} \quad (3)$$

with

$$\eta = \eta_0 + \eta_1(t - \bar{t}) + \eta_2(EHB_{it} - \overline{EHB_t}) \quad (4)$$

Original Paper

Outcome	Average risk sharing	Trend	Equity Home Bias
GNI	2 (1.02)	0 (0.02)	−39 (4.19)
Consumption	57 (15.06)	2 (2.09)	−136 (5.48)

Notes: t-values are reportet in parantheses.

Replication

Outcome	Average risk sharing	Trend	Equity Home Bias
GNI	15.67 (4.16)	0.75 (0.76)	−132.83 (4.87)
Consumption	54.56 (15.01)	0.55 (0.52)	−204.217 (9.07)

Notes: t-values are reportet in parantheses.

Risk Sharing and Foreign Asset Holdings relative to GDP, OECD Sample 1993–2003

Original Paper

Outcome	Measure	Equity	Debt	FDI	Equity + Debt	All Assets
Income	Average risk sharing	6 (2.74)	5 (2.53)	3 (1.70)	6 (2.85)	6 (2.75)
	Trend	0 (0.77)	0 (0.34)	0 (0.47)	0 (0.37)	0 (0.47)
	Asset holdings	5 (4.50)	9 (4.46)	3 (2.70)	9 (4.65)	8 (4.35)
Consumption	Average risk sharing	51 (13.34)	45 (11.56)	51 (13.39)	47 (11.95)	48 (12.34)
	Trend	1 (0.83)	0 (0.25)	0 (0.09)	0 (0.33)	0 (0.45)
	Asset holdings	11 (4.02)	7 (1.64)	9 (3.79)	9 (2.26)	11 (2.74)

Notes: t-values are reportet in parantheses.

Replication

Outcome	Measure	Equity	Debt	FDI	Equity + Debt	All Assets
Income	Average risk sharing	15.064 (4.145)	15.784 (4.211)	9.675 (2.788)	15.997 (4.253)	15.709 (4.262)
	Trend	1.202 (1.209)	0.715 (0.719)	1.685 (1.634)	0.733 (0.729)	1.069 (1.062)
	Asset holdings	12.180 (5.190)	17.747 (5.154)	11.726 (3.333)	17.583 (5.046)	20.515 (5.142)
Consumption	Average risk sharing	56.614 (14.769)	49.948 (12.157)	51.790 (12.591)	52.469 (12.928)	54.959 (13.589)
	Trend	0.942 (0.903)	0.172 (0.153)	0.455 (0.407)	0.355 (0.322)	0.497 (0.457)
	Asset holdings	17.834 (8.796)	21.312 (6.009)	14.252 (6.160)	22.576 (6.814)	24.189 (7.348)

Notes:t-values are reportet in parantheses.

Extention by time Risk Sharing and Equity Home Bias

Outcome	Average risk sharing	Trend	Equity Home Bias
GNI	0.24 (0.14)	−0.15 (0.82)	−28.32 (2.80)
Consumption	40.32 (19.61)	−0.07 (0.29)	−81.26 (7.96)

Notes:t-values are reportet in parantheses.

Extention by time and sample size Risk Sharing and Equity Home Bias

Outcome	Average risk sharing	Trend	Equity Home Bias
GNI	0.48 (0.34)	−0.04 (0.26)	−28.34 (3.24)
Consumption	30.39 (15.28)	−0.09 (0.41)	−84.31 (8.25)

Notes:t-values are reportet in parantheses.

Risk Sharing and Foreign Asset Holdings Relative to GDP: OECD 1987–2017

Time Extension Risk Sharing and Foreign Asset Holdings Relative to GDP

Outcome	Asset measure	Average risk sharing	Trend	Asset holdings
GNI	Equity	−1.70 (0.87)	−0.12 (0.70)	4.84 (3.59)
	Debt	−0.36 (0.21)	−0.14 (0.80)	7.60 (4.14)
	FDI	−2.31 (0.98)	0.07 (0.40)	4.39 (2.57)
	Equity+Debt	−1.13 (0.62)	−0.16 (0.91)	7.93 (4.15)
	All Assets	−1.97 (1.03)	−0.13 (0.74)	8.16 (4.08)
Consumption	Equity	42.22 (20.31)	0.06 (0.28)	10.50 (9.07)
	Debt	38.62 (18.10)	−0.17 (0.72)	11.94 (5.60)
	FDI	40.24 (19.21)	0.04 (0.18)	10.71 (7.41)
	Equity+Debt	40.26 (18.98)	−0.14 (0.58)	14.25 (7.06)
	All Assets	41.07 (19.35)	−0.09 (0.40)	14.97 (7.60)

Notes: t-values are reportet in parantheses.

Time and Sample Extension Risk Sharing and Foreign Asset Holdings Relative to GDP

Outcome	Asset measure	Average risk sharing	Trend	Asset holdings
GNI	Equity	0.89 (0.63)	−0.10 (0.68)	2.51 (3.40)
	Debt	0.28 (0.20)	−0.13 (0.93)	7.22 (4.72)
	FDI	−2.09 (1.23)	0.08 (0.62)	4.44 (3.23)
	Equity+Debt	0.07 (0.05)	−0.16 (1.09)	6.96 (4.65)
	All Assets	−0.38 (0.27)	−0.14 (0.94)	6.83 (4.54)
Consumption	Equity	27.96 (14.42)	−0.12 (0.56)	11.92 (11.26)
	Debt	29.72 (14.02)	0.11 (0.49)	9.69 (4.75)
	FDI	25.92 (11.85)	0.44 (2.00)	11.75 (7.78)
	Equity+Debt	28.45 (13.53)	0.0005 (0.002)	13.18 (6.73)
	All Assets	28.15 (13.46)	0.06 (0.27)	14.23 (7.37)

Notes: t-values are reportet in parantheses.